monitor the position, direction, orientation, velocity, acceleration and spin of the helicopter in several dimensions. A ground-based computer crunches the data, makes quick calculations and beams new flight directions to the helicopter via radio 20 times per second.

The helicopter carries accelerometers, gyroscopes and magnetometers, the latter of which use the Earth's magnetic field to figure out which way the helicopter is pointed. The exact location of the craft is tracked either by a GPS receiver on the helicopter or by cameras on the ground. (With a larger helicopter, the entire navigation package could be airborne.)

There's interest in using autonomous helicopters to search for land mines in war-torn areas or to map out the hot spots of wildfires, allowing firefighters to quickly move toward or away from them. Firefighters now must often act on information that is several hours old, Abbeel said.

"In order for us to trust helicopters in these sort of mission-critical applications, it's important that we have very robust, very reliable helicopter controllers that can fly maybe as well as the best human pilots in the world can," Ng said. Stanford's autonomous helicopters have taken a large step in that direction, he said.

Chess and AI

We often think that artificial intelligence is at its peak in the field of video games, since you're forever playing against something called the "AI" of the game. But in reality, games use a weak AI that's more about a set of instructions given according to the situation the player finds himself in. This is why you'll often find a "trick" to beating the video game bots.

Still, virtual games do play a significant role in demonstrating artificial intelligence. Computers have already come so far that they've "solved" checkers. The biggest field of advance now is chess-playing computers Deep Blue and Fritz. Given that chess is often an indication of a human's intelligence, it's no surprise then that the ability of a computer to defeat a human is considered one of the finest achievements in AI.

Computers and chess have a deep history. In 1968, international master David Levy made a famous challenge saying that no computer could beat him in the next 10 years. In 1978, he did manage to defeat